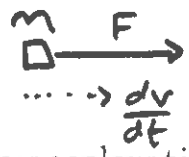
Lecture 4

Separable and Linear 1st-Order Equations

In many circumstances we can model physical behavior with differential equations. Newton's second law of motion states that the total force acting on an object is equal to its mass times its acceleration. This can be expressed as

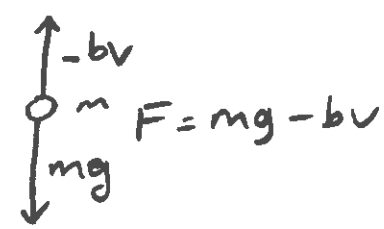
$$m\frac{dv}{dt} = F$$



where F is the total force, m is the mass, and $\frac{dv}{dt}$ is the acceleration (derivative of velocity).

When an object is near the surface of the Earth, gravity imposes a force on the object which can be expressed as mg where g is the acceleration due to gravity. If the object is falling towards the Earth's surface it is acted upon by gravity but also some form of air resistance. We can reasonably model this air resistance as an opposing force $-bv$ where b is a constant that depends upon the object. In this circumstance, the total force can be written as

$$m\frac{dv}{dt} = mg - bv$$



We can solve this for $\frac{dv}{dt}$ to obtain

$$\boxed{\frac{dv}{dt} = \frac{mg - bv}{m} = f(t,v)}$$

Notice that this equation is of the form $\frac{dv}{dt} = f(t, v)$ which is one of the forms of a 1st-order equation we have been studying. In this lecture, we will con-

sider such 1st-order equations and begin our study of solving certain types of differential equations analytically.

We have been considering the 1st-order equation of the form

$$\frac{dy}{dx} = f(x, y).$$

If we have a bit more information about the makeup of $f(x, y)$ we can find solutions to such equations.

Definition: A 1st-order differential equation is **separable** if it can be written in the form

$$\frac{dy}{dx} = \underbrace{g(x)}_{\text{factor apart}}\underbrace{p(y)}_{} \quad (= f(x,y)) \tag{4.1}$$

Example: Are the following equations separable or not separable?

1. $\dfrac{dy}{dx} = x - x^2y = x(1 - xy)$ not separable

2. $\dfrac{dy}{dx} = \dfrac{ye^{x+y}}{x^2+2} = \dfrac{y(e^x \cdot e^y)}{x^2+2} = \underbrace{(ye^y)}_{p(y)}\underbrace{\left(\dfrac{e^x}{x^2+2}\right)}_{g(x)}$ separable

In general, we can solve any separable equation using the following method:

1. Write the equation as

$$\frac{dy}{dx} = g(x)p(y) \leftarrow \text{factor}$$

2. Treat dy and dx as differentials and multiply by dx to obtain

$$dy = (g(x)dx)p(y)$$

3. Divide by $p(y)$ (What if $p(y) = 0$?) ? extraneous possible solutions

$$\frac{1}{p(y)}\,dy = \frac{dy}{p(y)} = g(x)dx$$

4. Write $\dfrac{1}{p(y)} = h(y)$ to obtain

$$h(y)dy = g(x)dx \Rightarrow \int h(y)dy = \int g(x)dx + C$$

5. Integrate both sides to obtain an implicit solution

$$H(y) = G(x) + C$$

NOTE: We must check the zeros of $p(y)$ prior to division and see if they are solutions to the equation.

Example: Consider the separable equation $\dfrac{dy}{dx} = \underbrace{(x-2)^2}_{g(x)}\underbrace{(y-13)}_{p(y)}$. separable

Here $p(y) = y - 13$ $(= 0)$

$\Rightarrow y - 13 = 0 \Rightarrow y = 13 \leftarrow$ Function

Is $y = 13$ a solution? $\dfrac{dy}{dx} = \dfrac{d}{dx}(13) = 0$, so $0 \overset{?}{=} (x-2)^2(13-13)$

$\Rightarrow 0 = (x-2)^2(0)$

$\Rightarrow 0 = 0$ ✓ yes!

To solve the rest: use the method

(1) $\dfrac{dy}{dx} = (x-2)^2(y-13)$

(2) $dy = (x-2)^2 dx\,(y-13)$ multiply by dx

(3) $\dfrac{1}{y-13}\,dy = (x-2)^2 dx$ divide by $y-13$

$u = x-2,\ du = dx$

④ & ⑤ $\displaystyle\int \frac{1}{y-13}\,dy = \int (x-2)^2\,dx + C$ ← pos or neg

$u = y-13$, $du = dy$

$\Rightarrow \ln|y-13| = \dfrac{(x-2)^3}{3} + C$ (implicit solution)

$\Rightarrow |y-13| = e^{\frac{(x-2)^3}{3}+C} = e^{\frac{(x-2)^3}{3}}\, e^{C}$ $= C$ ← pos

$\Rightarrow y-13 = e^{\frac{1}{3}(x-2)^3}\cdot \underset{\uparrow \pm}{C} \Rightarrow \boxed{y = Ce^{1/3(x-2)^3} + \underline{\underline{13}}}$ → $C=0$

Example: Recall the motion of an object falling towards Earth's surface with air resistance is given by

$$\frac{dv}{dt} = \frac{mg - bv}{m}$$

$\uparrow -bv$, $\downarrow mg$, $F = mg - bv$

which is separable. Solve the equation and explain the asymptotic behavior of $v(t)$ as $t \to \infty$.

t ind v dep

$p(v) = 0$?

$$\frac{dv}{dt} = \underbrace{\frac{mg-bv}{m}}_{p(v)} \cdot \underbrace{1}_{g(t)} \quad \text{separable}$$

$\Rightarrow \dfrac{mg-bv}{m} = 0$

$\Rightarrow mg - bv = 0$

$\Rightarrow bv = mg$

$\Rightarrow v = \dfrac{mg}{b}$

Is $v = \dfrac{mg}{b}$ a solution? yes!

$v' = 0$ $\qquad 0 \overset{?}{=} \dfrac{mg - b\left(\frac{mg}{b}\right)}{m}$ ✓

Solve with the method:

$$dv = \frac{mg-bv}{m}\,dt$$

divide by $mg-bv$ $\Rightarrow \displaystyle\int \frac{1}{mg-bv}\,dv = \int \frac{1}{m}\,dt + C$

$u = mg-bv$, $du = -b\,dv$

$\Rightarrow -\dfrac{1}{b}\ln|mg-bv| = \dfrac{1}{m}t + C$

$\Rightarrow \ln|mg-bv| = -\dfrac{b}{m}t + C$ ← $-bC$

$mg - bv = e^{-\frac{b}{m}t + \underline{\underline{C}}} \Rightarrow \boxed{v = \dfrac{mg}{b} - Ce^{-\frac{b}{m}t}}$

$\displaystyle\lim_{t\to\infty} v(t) = \lim_{t\to\infty}\left(\frac{mg}{b} - Ce^{-\frac{b}{m}t}\right)$ → 0

$\lim_{t\to-\infty} e^{t} = 0$

$= \dfrac{mg}{b} - 0$

$= \dfrac{mg}{b}$ terminal velocity

Example: Consider the IVP

$$\frac{dy}{dx} = \frac{y+1}{x-3}, \quad y(0) = 1$$

$$\frac{dy}{dx} = \frac{y+1}{x-3} = \underbrace{\frac{1}{x-3}}_{g(x)} \cdot \underbrace{(y+1)}_{p(y)}$$

$p(y) = 0$

$\Rightarrow y+1 = 0$

$\Rightarrow y = -1$

particular solution?

No, does not satisfy

~~$y(0)=1$~~, $y(0) = -1$

Method: $\frac{1}{y+1}\,dy = \frac{1}{x-3}\,dx$

$$\Rightarrow \int \frac{1}{y+1}\,dy = \int \frac{1}{x-3}\,dx + C$$

$$\Rightarrow \ln|y+1| = \ln|x-3| + C$$

$$\Rightarrow y+1 = C(x-3) = C(x-3)$$

contains 1 for $|y+1|$

$$\Rightarrow \Rightarrow y = C(x-3) - 1$$ lines.

If $y(0) = 1 \Rightarrow 1 = y(0) = C(0-3) - 1$

$\Rightarrow 1 = -3C - 1 \Rightarrow -3C = 2$

$\Rightarrow C = -2/3$

The particular solution is $\boxed{y = -\frac{2}{3}(x-3) - 1}$

Example: Solve the IVP

$$\frac{dy}{dx} = \frac{y^2-1}{x^2-1}, \quad y(2) = 2$$

$$\frac{dy}{dx} = \frac{1}{x^2-1}(y^2-1)$$

$y(2)=2.$

$p(y)=0$

$\Rightarrow y^2-1=0$

$\Rightarrow y=\pm 1$ X not particular solution.

Method:

$$\frac{1}{y^2-1}\,dy = \frac{1}{x^2-1}\,dx \Rightarrow \int \frac{1}{y^2-1}\,dy = \int \frac{1}{x^2-1}\,dx + C$$

$$\Rightarrow \int \frac{1}{(y-1)(y+1)}\,dy = \int \frac{1}{(x-1)(x+1)}\,dx + C$$

PFD: $\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} \Rightarrow 1 = A(x+1) + B(x-1)$

$x=1: \quad 1 = A(1+1) + B(0) \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$

$x=-1: \quad 1 = A(0) + B(-1-1) \Rightarrow 1 = -2B \Rightarrow B = -1/2$

So $\frac{1}{(x-1)(x+1)} = \frac{1/2}{x-1} + \frac{-1/2}{x+1}$

$$\Rightarrow \int\left(\frac{1/2}{y-1} - \frac{1/2}{y+1}\right)dy = \int\left(\frac{1/2}{x-1} - \frac{1/2}{x+1}\right)dx + C$$

$\Rightarrow \frac{1}{2}\ln|y-1| - \frac{1}{2}\ln|y+1| = \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$ multiply by 2

$\Rightarrow \ln\left|\frac{y-1}{y+1}\right| = \ln\left|\frac{x-1}{x+1}\right| + C \Rightarrow \frac{y-1}{y+1} = C\left(\frac{x-1}{x+1}\right)$ implicit solution.

IVP

$y(2)=2$, $(2,2)$ $\boxed{\frac{2-1}{2+1} = C\left(\frac{2-1}{2+1}\right)} \Rightarrow C=1$ so $\boxed{\frac{y-1}{y+1} = \frac{x-1}{x+1}} \Rightarrow \dots \Rightarrow \boxed{y=x}$

Not every 1st-order equation can be written as a separable equation. Recall, a linear 1st-order equation is of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \quad a_1(x) \neq 0 \tag{4.2}$$

Since $a_1(x) \neq 0$ we usually rewrite the equation by dividing through by a_1.

Definition: A linear 1st-order equation is in **standard form** if it is written as

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)} \tag{4.3}$$

where $P(x) = \dfrac{a_0}{a_1}$ and $Q(x) = \dfrac{g}{a_1}$.

There are two cases to consider when solving a linear 1st order equation in standard form.

Case 1: If $P(x) = 0$, then we have $\dfrac{dy}{dx} = Q(x)$. This is a separable equation and can immediately be solved by

$$y(x) = \int Q(x)dx + C$$

Case 2: If $P(x) \neq 0$, we consider the equation

? Product Rule

Product Rule:

$(fg)' = f'g + fg'$

$$\left(\frac{dy}{dx} + P(x)y\right) = Q(x).$$

Notice how the left hand side (LHS) contains one term with $\dfrac{dy}{dx}$ and one term with y. If we look at the situation where we have a function $\mu(x)$ times y and differentiate using the product rule, we have

$$\frac{d}{dx}(\mu(x)y) = \underbrace{\mu(x)\frac{dy}{dx} + \mu'(x)y}. \quad \longleftrightarrow \frac{dy}{dx} + P(x)y$$

This is similar in form to the LHS of (4.3). We are missing the $\mu(x)$ as a coefficient of $\frac{dy}{dx}$. If we multiply (4.3) by a function $\mu(x)$ we obtain

$$\boxed{\mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)}$$

what $\mu(x)$ gives us the product rule?

If the LHS is of the form $\frac{d}{dx}(\mu(x)y)$ then we can solve for y by integrating both sides with respect to x and then dividing by $\mu(x)$.

In general, not every linear 1st-order equation is of this form. However, we can force every linear 1st-order equation into this form with the following method:

1. Given a linear 1st-order equation of the form (4.3), multiply both sides by $\mu(x)$ to obtain
$$\boxed{\mu(x)\frac{dy}{dx} + \boxed{\mu(x)P(x)}y} = \mu(x)Q(x)$$
same!
2. To have the LHS be of the form
$$\frac{d}{dx}(\mu(x)y) = \boxed{\mu(x)\frac{dy}{dx} + \boxed{\mu'(x)}y}$$
we need $\mu(x)P(x)y = \mu'(x)y$ or
$$\boxed{\mu'(x) = \mu(x)P(x).} \rightarrow \frac{d\mu}{dx} = \underbrace{\mu(x)}_{p(\mu)}\underbrace{P(x)}_{q(x)}$$
3. This is a separable equation where μ is the dependent variable and x is the independent variable. We can solve this and obtain

$$\frac{d\mu}{dx} = \mu \cdot P(x) \Rightarrow \int \frac{1}{\mu}\, d\mu = \int P(x)\,dx \cancel{+C} \quad \text{no } +C.$$

$$\Rightarrow \ln|\mu| = \int P(x)\,dx \cancel{+C} \Rightarrow$$

$$\boxed{\mu(x) = e^{\int P(x)\,dx}}$$

4. Therefore we can rewrite (4.3) as

$$\mu(x)\frac{dy}{dx} + \overset{\mu(x)P(x)}{\mu' y} = \int \frac{d}{dx}(\mu(x)y) = \int \mu(x)Q(x). + C$$

Integrating with respect to x we obtain

$$\mu(x)y = \int \mu(x)Q(x)dx + C \Rightarrow \quad y = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)dx + C\right]$$

This is summarized in the theorem below:

Theorem (Integrating Factors for Linear 1st-Order Equations): Given a linear 1st-order equation in standard form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

we use

$$\mu(x) = e^{\int P(x)dx}$$

called an **integrating factor** for the equation to write the general solution as

$$y = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)dx + C\right]$$

NOTE: You can memorize the general solution form with the integrating factor or can use the method given before to construct the solution using the integrating factor.

Example: Solve $(x^2+9)\frac{dy}{dx} + xy = 0$ no y's linear? yes ✓

<u>Put in standard form:</u>

divide by x^2+9:

$$\frac{dy}{dx} + \frac{x}{x^2+9}y = 0 \qquad (*)$$

$u = x^2+9$
$du = 2x\,dx$

$P(x) = \frac{x}{x^2+9}$, $Q(x) = 0$.

integrating factor: $\mu(x) = e^{\int P(x)dx} = e^{\int \frac{x}{x^2+9}dx}$

$= e^{\frac{1}{2}\ln|x^2+9|}$

$= e^{\ln((x^2+9)^{1/2})}$

$= \sqrt{x^2+9}$

$\frac{x}{(\sqrt{x^2+9})^{\cancel{2}}} \cdot \sqrt{\cancel{x^2+9}}$

So if we multiply $(*)$ by $\mu(x) = \sqrt{x^2+9}$, we get

$$\underbrace{\sqrt{x^2+9}\,\frac{dy}{dx} + \frac{x}{\sqrt{x^2+9}}y}_{\text{product rule!}} = 0$$

$$\Rightarrow \frac{d}{dx}\underbrace{\left(\sqrt{x^2+9}\,y\right)}_{\mu\cdot y} = \underbrace{0}_{\mu\cdot Q} \Rightarrow \sqrt{x^2+9}\,y = \int 0\,dx + C$$

$$\Rightarrow \sqrt{x^2+9}\,y = C$$

$$\Rightarrow \boxed{y = \frac{C}{\sqrt{x^2+9}}}$$

Example: Solve $x\frac{dy}{dx} - 4y = x^7e^x$

power 1

linear!

no y's

!!!

Put in standard form: divide by x

$$\frac{dy}{dx} - \frac{4}{x}y = x^6e^x$$

$$\begin{cases} P(x) = -\frac{4}{x} \\ Q(x) = x^6e^x \end{cases}$$

integrating factor: $\mu(x) = e^{\int P(x)dx}$

$$= e^{\int -\frac{4}{x}dx} = e^{-4\ln|x|} = e^{\ln(x^{-4})}$$

$$= x^{-4}$$

So $y = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)dx + C\right]$

$$= \frac{1}{x^{-4}}\left[\int x^{-4}\cdot x^6e^x dx + C\right]$$

$$= x^4\left[\int x^2e^x dx + C\right]$$

IBP: Table method

d		$\int$
x^2	$+$	e^x
$2x$	$-$	e^x
2	$+$	e^x
0		e^x

so $\int x^2e^x dx$

$= x^2e^x - 2xe^x + 2e^x$

$$= x^4\left[(x^2e^x - 2xe^x + 2e^x) + C\right]$$

$$\Rightarrow \boxed{y = x^6e^x - 2x^5e^x + 2x^4e^x + Cx^4}$$

Example: Consider the IVP

y dep
x ind

$$\frac{dy}{dx} = \frac{1}{x + y^2}, \quad y(-2) = 0$$

$(-2, 0)$
x y

not linear! $\underline{y^2}$

Shift to x dep, y ind

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\frac{1}{x+y^2}} = x + y^2$$

$$\Rightarrow \frac{dx}{dy} - x = y^2 \quad \text{linear for x dep.}$$

$$P(y) = -1, \quad Q(y) = y^2, \quad \mu(y) = e^{\int P(y)dy}$$

$$\Rightarrow \mu(y) = e^{\int -1 dy} = e^{-y}$$

So $x = \frac{1}{\mu(y)}\left[\int \mu(y) Q(y)\, dy + C\right]$

IBP:

y^2	$+$	e^{-y}
$2y$	$-$	$-e^{-y}$
2	$+$	e^{-y}
0		$-e^{-y}$

$$= \frac{1}{e^{-y}}\left[\int e^{-y} y^2\, dy + C\right]$$

$$= \frac{1}{e^{-y}}\left[-y^2 e^{-y} - 2y e^{-y} - 2e^{-y} + C\right]$$

$$\Rightarrow \boxed{x = -y^2 - 2y - 2 + Ce^{y}}$$

explicit if x dep
implicit if y dep

For $(-2, 0)$ $\quad -2 = 0 - 0 - 2 + Ce^0 \Rightarrow -2 = -2 + C \Rightarrow C = 0$
x y

$$\boxed{x = -y^2 - 2y - 2}$$